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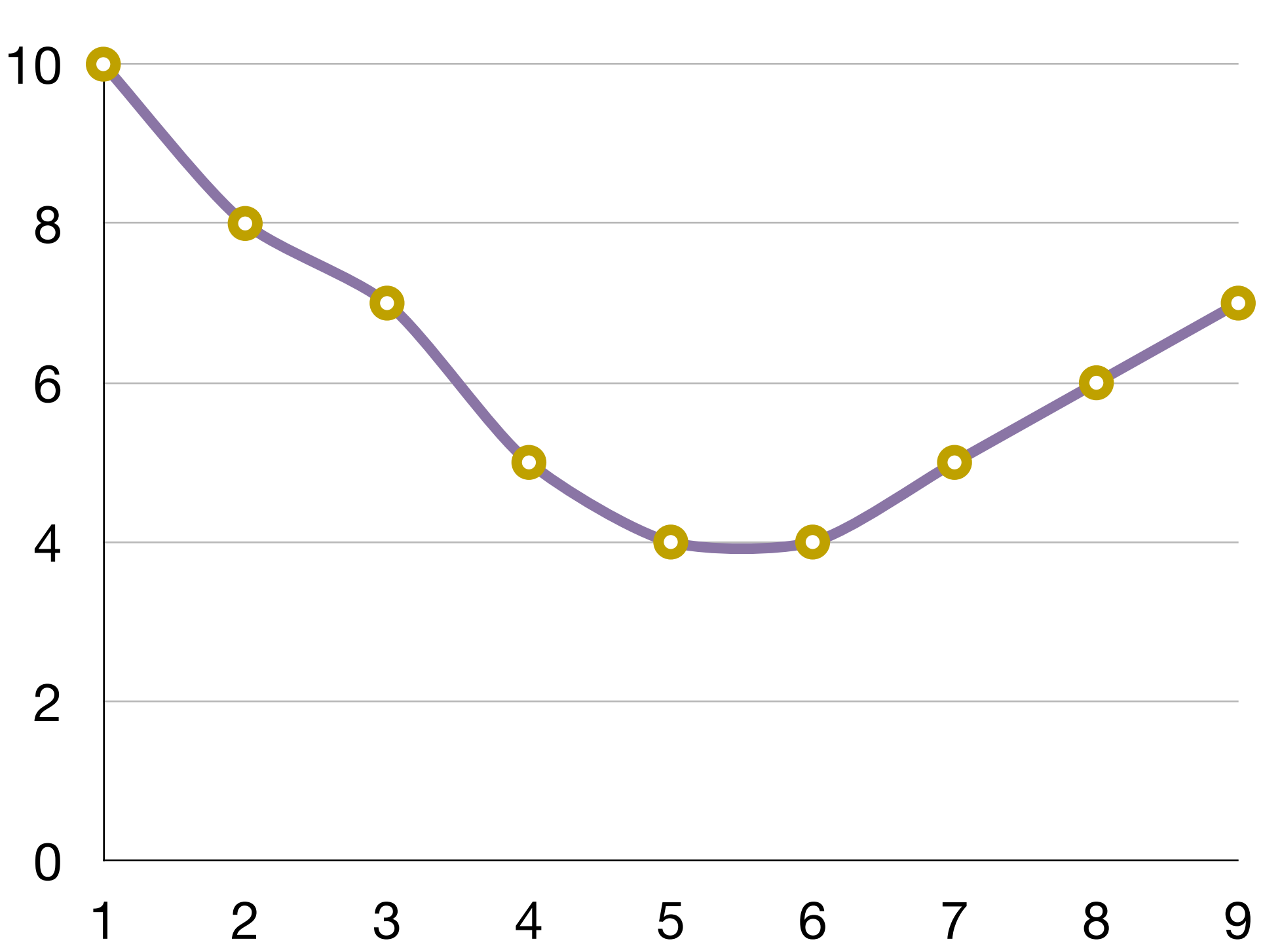


Institute of Health Policy, Management and Evaluation
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SELECTING THE OPTIMAL *K* IN KNN







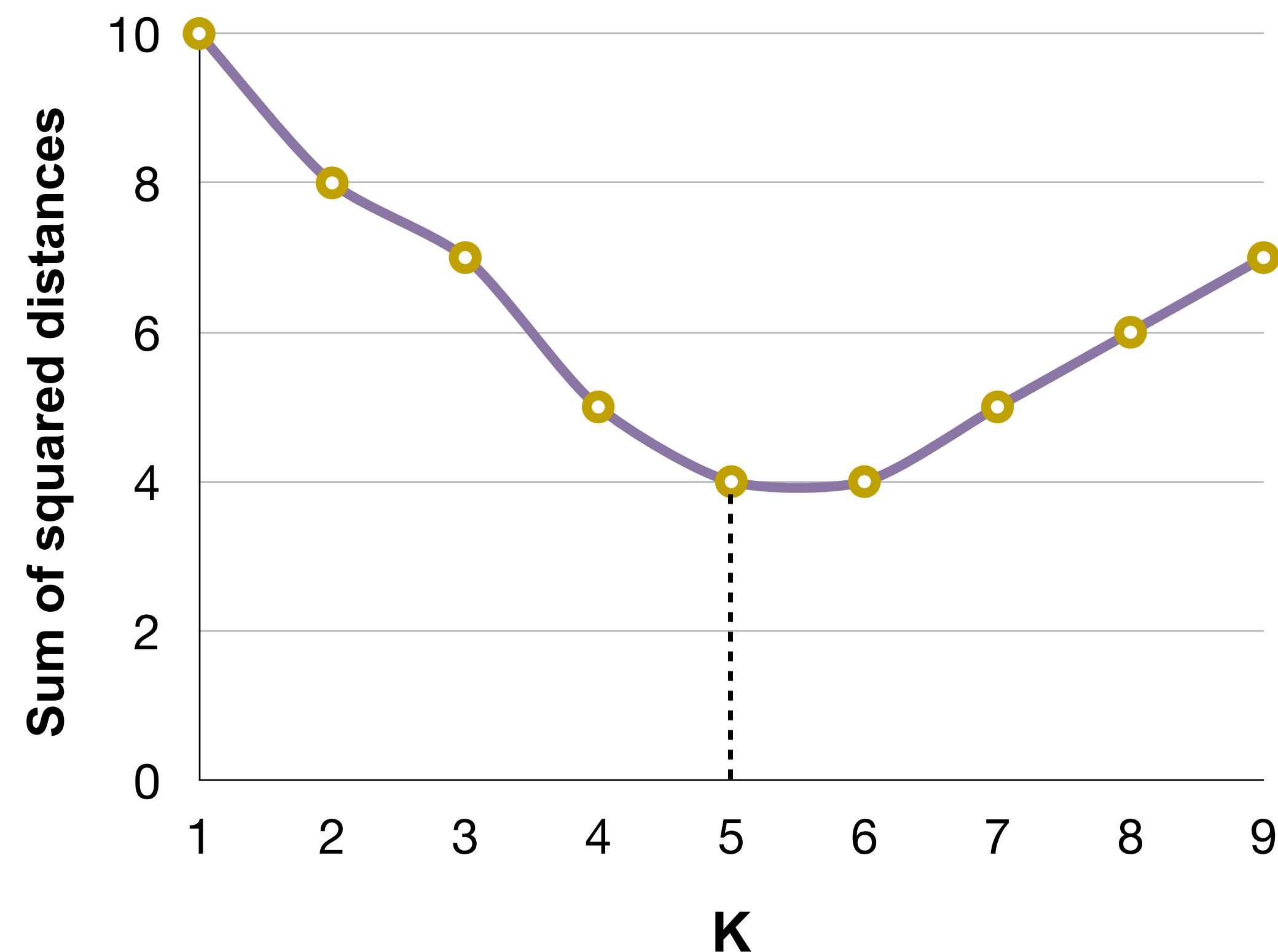
- ▶ Compute the sum of squared distances from each point to its assigned center.
- ▶ Plot these values for a range of K values.
- ▶ Look for an "elbow" in the plot, a point where the rate of decrease sharply changes. This can indicate a good estimate for K .

The idea is that, after a certain point, adding more clusters will not provide much better fit to the data.

Elbow Method



SELECTING THE OPTIMAL K IN KNN



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HANDLING OUTLIERS

In a dataset where you are clustering health metrics of individuals, how might outliers impact **K-means** clustering?

